CHAPTER FOUR

DATA ANALYSIS, INTERPRETATION AND DISCUSSION OF FINDINGS

4.1 Introduction

This section provides an overview of the analysis and the variables employed in examining the impact of Corporate Social Responsibility Disclosure (CSRD) on the financial performance of Nigerian oil and gas companies. The study focuses on the period from 2014 to 2023, applying panel data regression techniques to evaluate the relationship between selected CSR disclosure components and financial performance.

To achieve this objective, the study utilizes multiple panel regression models, including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects estimators. Furthermore, the Hausman specification test is employed to determine the most appropriate model for the data.

The analysis is centred on the following key variables:

- **ROA (Return on Assets):** This is the dependent variable used to measure a company's profitability relative to its total assets.
- **Community Relation and Sustenance Disclosure (CRSD):** A binary independent variable representing whether the firm publicly discloses its efforts to maintain positive relations with host communities.
- **Community Investment Disclosure (CID):** A binary independent variable indicating the disclosure of investments made toward community development, such as education, infrastructure, and public health.

- **Health and Safety Program Disclosure (HSPD):** A binary independent variable capturing whether the firm discloses information related to workplace safety and employee well-being.
- **Firm Size (FS):** A control variable measured as the natural logarithm of total assets, included to account for firm-level size differences in financial performance.

Through this analytical framework, the study aims to determine whether and to what extent CSR disclosures—particularly in the areas of community engagement, investment, and safety—affect the financial performance of listed oil and gas companies in Nigeria.

4.2 Presentation and Interpretation of Results

4.2.1 Descriptive Statistics

The first step in the analysis is to examine the basic statistical properties of the variables using descriptive statistics. Table 4.1 presents the mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and the Jarque-Bera normality **test** for each variable.

Table 4.1: Descriptive Statistics

	ROA____	CRSD	CID	HSPD	FS
Mean	7.007111	0.711111	0.855556	0.755556	5.725499
Median	7.010000	1.000000	1.000000	1.000000	5.632250
Maximum	10.08000	1.000000	1.000000	1.000000	9.472700
Minimum	4.300000	0.000000	0.000000	0.000000	2.708100
Std. Dev.	1.123146	0.455785	0.353509	0.432165	1.938801
Skewness	0.142278	-0.931552	-2.022847	-1.189302	0.337497
Kurtosis	3.262027	1.867788	5.091908	2.414439	1.871393

Jarque-Bera	0.561112	17.82396	77.78892	22.50239	6.485142
Probability	0.755364	0.000135	0.000000	0.000013	0.039063
Sum	630.6400	64.00000	77.00000	68.00000	515.2949
Sum Sq. Dev.	112.2696	18.48889	11.12222	16.62222	334.5465
Observations	90	90	90	90	90

Source: Author's computation using Eview 9

Table 4.1 presents the descriptive statistics for the study variables based on 90 firm-year observations. The dependent variable, Return on Assets (ROA), has a mean value of 7.01%, indicating moderate profitability among the sampled firms, with values ranging from 4.30% to 10.08%. The low standard deviation (1.12) and nearly symmetrical distribution (skewness = 0.14) suggest consistent financial performance across firms.

Among the CSR disclosure variables, Community Investment Disclosure (CID) had the highest average score (0.86), followed by Health and Safety Program Disclosure (HSPD) (0.76) and Community Relation and Sustenance Disclosure (CRSD) (0.71), reflecting generally high levels of CSR reporting. These variables are binary, hence their standard deviations are relatively low and their distributions are not normal, as confirmed by significant Jarque-Bera test results. The control variable, Firm Size (FS), showed considerable variation (mean = 5.73, std. dev. = 1.94), indicating the sample includes both small and large firms. Overall, the data suggests that CSR disclosures are widely practiced, particularly in community investment and health-related areas.

4.2.2 Correlation matrix

The next step in the analysis involves examining the relationships between the variables using a correlation matrix. The correlation matrix provides insights into the strength and direction of linear relationships between the variables. It helps identify potential multicollinearity issues and offers a

preliminary understanding of how CSR disclosures may relate to financial performance indicator such as ROA. The matrix includes Pearson correlation coefficients for each pair of variables, which are essential for assessing the degree of association between them.

Table 4.2: Correlation matrix

	ROA____	CRSD	CID	HSPD	FS
ROA_ ___	1	0.01393513364 360273	0.02695334601 967901	- 0.04637945756 45686	0.22894101305 93238
CRSD	0.01393513364 360273	1	0.15651561416 64682	- 0.02028184785 787094	- 0.04006660173 884487
CID	0.02695334601 967901	0.15651561416 64682	1	0.06047124139 994298	- 0.02424318346 449164
HSPD	- 0.04637945756 45686	- 0.02028184785 787094	0.06047124139 994298	1	0.02600425009 060487
FS	0.22894101305 93238	- 0.04006660173 884487	- 0.02424318346 449164	0.02600425009 060487	1

Source: Author's computation using Eview 9

Table 4.2 presents the correlation matrix that examines the pairwise linear relationships among the variables used in the study: Return on Assets (ROA), Corporate Social Responsibility Disclosure (CRSD), Community Involvement Disclosure (CID), Health and Safety Practices Disclosure (HSPD), and Firm Size (FS). The correlation coefficients provide preliminary insights into the direction and strength of associations between the variables before conducting multivariate regression analysis.

The correlation between ROA and CRSD is weak and positive (r = 0.01), suggesting a negligible linear relationship between overall CSR disclosure and firm profitability. Similarly, the correlation between ROA and CID is weak (r = 0.03), indicating that community engagement disclosure is not strongly associated with profitability. Interestingly, HSPD exhibits a weak negative correlation with ROA (r = -0.05), implying that increased health and safety disclosures may slightly coincide with reduced profitability. This could reflect the cost-intensive nature of health and safety investments, which may not yield immediate financial returns.

Firm Size (FS) shows the strongest correlation with ROA (r = 0.23) among all independent variables, though still modest. This suggests that larger firms, possibly due to economies of scale, more efficient resource utilization, or greater market power, tend to report higher profitability. This finding aligns with prior literature which identifies firm size as a potential determinant of financial performance.

Among the independent variables, CRSD and CID exhibit a low-to-moderate positive correlation (r = 0.16), which is expected since community involvement is often a component of broader CSR practices. Nonetheless, the strength of this relationship remains weak, underscoring the multidimensional and sometimes fragmented nature of CSR activities. Other pairwise correlations such as CRSD-HSPD (r = -0.02), CID-HSPD (r = 0.06), CRSD-FS (r = -0.04), and CID-FS (r = -0.02) are negligible, indicating that these dimensions of CSR disclosure are largely independent of one another and of firm size.

Importantly, none of the correlation coefficients exceed the threshold of 0.7, which is often considered a warning sign for multicollinearity in regression analysis. This suggests that the explanatory variables do not exhibit strong intercorrelations that could distort the regression estimates or inflate standard errors. Therefore, the dataset is well-suited for further inferential

analysis, including regression modelling, without the need for variable transformation or exclusion due to multicollinearity concerns.

In summary, the correlation analysis provides preliminary evidence that the linear associations among the study variables are generally weak. While Firm Size shows some association with financial performance, the direct relationships between CSR disclosure components and ROA appear minimal. These findings highlight the need for more robust multivariate analysis to ascertain the true nature and significance of these relationships, considering possible confounding variables and interaction effects.

4.2.3 Variance Inflation Factor (VIF) Analysis

To assess the potential issue of multicollinearity among the independent variables, a Variance Inflation Factor (VIF) analysis was conducted. The VIF values for the independent variables in the model are summarized in Table 4.3.

Table 4.3: Variance Inflation Factor (VIF)

Variable	VIF
CRSD	1.03
CID	1.03
HSPD	1.01
FS	1.00

As shown in Table 4.3, all VIF values fall significantly below the commonly used threshold of 5, and even the more conservative threshold of 2.5. CRSD and CID each have a VIF of 1.03, while HSPD and Firm Size exhibit even lower values of 1.01 and 1.00, respectively. These findings indicate a very low level of intercorrelation among the explanatory variables.

From a statistical standpoint, these VIF values confirm that the independent variables are not linearly dependent on one another to a degree that would threaten the validity of the regression results. This reinforces the earlier findings from the correlation matrix (Table 4.2), where all pairwise correlations among the independent variables were weak. The convergence of these two diagnostic tests (correlation matrix and VIF) enhances the robustness of the multicollinearity assessment.

The low VIF values also imply greater confidence in the stability and reliability of the regression coefficients that will be estimated in subsequent sections. Since multicollinearity inflates the standard errors of coefficients, its absence ensures that the model's statistical tests (e.g., t-tests and F-tests) are more trustworthy, and that the interpretation of each variable's unique contribution to the dependent variable (ROA) is not compromised.

In summary, the VIF analysis confirms that multicollinearity is not a concern in this study. The independent variables—Corporate Social Responsibility Disclosure (CRSD), Community Involvement Disclosure (CID), Health and Safety Practices Disclosure (HSPD), and Firm Size (FS)—are sufficiently distinct, validating their inclusion in the multiple regression model for further inferential analysis.

4.2.4 Regression Analysis

The regression analysis is the core method used to examine the impact of CSR disclosures on the financial performance of Nigerian oil and gas companies. This step involves applying panel regression models, including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models, to assess the relationship between the independent variables (CSRD, CID, HPSD, FS) and the dependent variable, ROA. The regression coefficients provide insights into the direction and

magnitude of the impact of CSR disclosures on financial performance. Additionally, the analysis includes tests such as the Hausman test to determine the most appropriate model for the data, helping to ensure that the results are both reliable and robust. The regression analysis offers a more detailed understanding of how CSR disclosures influence financial outcomes in the oil and gas sector.

Regression Analysis for Return on Assets (ROA)

To examine the effect of Corporate Social Responsibility Disclosures (CSRD) on financial performance, Return on Assets (ROA) was adopted as one of the key financial proxies. The regression analysis involved the use of multiple estimation techniques including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models. The aim was to determine the most appropriate model that best explains the data. In order to validate the model choice, the Hausman specification test was conducted to compare the Fixed Effects and Random Effects models.

Table 4.4: Regression Analysis

Dependent Variable: ROA____

Method: Panel Least Squares

Date: 04/30/25 Time: 11:52

Sample: 2014 2023

Periods included: 10

Cross-sections included: 9

Total panel (balanced) observations: 90

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.268260	1.377840	2.372016	0.0202

CRSD	0.194445	0.275611	0.705506	0.4826
CID	-0.097713	0.362889	-0.269265	0.7884
HSPD	-0.029271	0.279617	-0.104683	0.9169
FS	0.647331	0.224167	2.887724	0.0050

	Effects Specification
Cross-section fixed (dummy variables)	

R-squared	0.196936	Mean dependent var	7.007111
Adjusted R-squared	0.071783	S.D. dependent var	1.123146
S.E. of regression	1.082084	Akaike info criterion	3.128539
Sum squared resid	90.15974	Schwarz criterion	3.489623
Log likelihood	-127.7843	Hannan-Quinn criter.	3.274149
F-statistic	1.573562	Durbin-Watson stat	1.991334
Prob(F-statistic)	0.117259		

Source: Author's computation using Eview 9

Table 4.4 presents the results of the fixed effects panel regression, which evaluates the influence of Corporate Social Responsibility Disclosure (CSRD) components—Community Relation and Sustenance Disclosure (CRSD), Community Investment Disclosure (CID), and Health and Safety Program Disclosure (HSPD)—as well as Firm Size (FS) on Return on Assets (ROA), a key measure of firm profitability. The analysis is based on a balanced panel of 90 firm-year observations, covering nine listed oil and gas firms in Nigeria over a 10-year period (2014–2023).

Key Findings:

Among the explanatory variables, Firm Size (FS) emerges as the only variable with a positive and statistically significant effect on ROA ($\beta = 0.647$, $p = 0.005$). This finding suggests that larger firms tend to be more profitable, possibly due to their superior resource base, economies of scale, stronger market presence, and better access to capital. This result aligns with prior empirical evidence that associates firm size with better financial performance.

In contrast, the three CSRD variables—CRSD ($p = 0.48$), CID ($p = 0.79$), and HSPD ($p = 0.92$)—exhibit no statistically significant influence on ROA. Despite being theoretically important components of corporate social responsibility, these disclosures appear not to translate into measurable improvements in short-term profitability. Several explanations could account for this outcome:

- CSR activities may not be effectively communicated or perceived as value-adding by investors and stakeholders.
- The impact of CSR initiatives may be long-term or intangible, thus not immediately reflected in accounting-based profitability metrics like ROA.
- The disclosure practices themselves may be compliance-driven rather than strategic, thereby limiting their financial relevance.

Model Diagnostics:

The coefficient of determination ($R^2 = 0.197$) indicates that approximately 19.7% of the variance in ROA is explained by the model. However, the adjusted R^2, which accounts for the number of predictors, drops to 0.072, reflecting only modest explanatory power. This suggests that other

factors not included in the model may significantly influence firm profitability in the Nigerian oil and gas sector.

The F-statistic (1.57, $p = 0.117$) shows that the model is not statistically significant at the conventional 5% level, implying that the set of explanatory variables does not jointly predict ROA with high confidence. Nevertheless, the Durbin-Watson statistic (1.99) is close to the ideal value of 2.0, indicating the absence of serious autocorrelation among the residuals and enhancing the reliability of the regression results.

Implications:

The regression findings underscore the limited role of CSR disclosure in driving short-term financial performance, at least within the context and timeframe of this study. While CSR may yield other strategic benefits—such as enhanced corporate image, stakeholder trust, and long-term sustainability—these advantages may not directly manifest in immediate profitability gains as measured by ROA.

The significant role of firm size highlights the importance of scale and operational capacity in achieving financial success, especially in capital-intensive sectors like oil and gas. This result calls for further research into how structural and operational factors mediate the relationship between CSR and performance.

Conclusion:

In sum, the fixed effects regression model suggests that while firm size positively influences profitability, the impact of CSR disclosures remains statistically insignificant. These results point to the need for more strategic and substantive CSR practices, rather than merely symbolic or

compliance-oriented disclosures, if firms wish to translate social responsibility into financial performance.

4.3 Discussion of Findings

The findings from this study offer meaningful insights into the dynamics between Corporate Social Responsibility Disclosure (CSRD) and the financial performance of listed Nigerian oil and gas firms over a ten-year period (2014–2023). Using a fixed effects panel regression model based on 90 firm-year observations, the analysis reveals that among the variables examined, Firm Size (FS) is the only statistically significant predictor of financial performance, as measured by Return on Assets (ROA).

In contrast, the three dimensions of CSR disclosure—Community Relation and Sustenance Disclosure (CRSD), Community Investment Disclosure (CID), and Health and Safety Program Disclosure (HSPD)—did not demonstrate statistically significant effects on firm profitability. These results shed light on the complex and often indirect nature of the CSR–performance nexus in the Nigerian oil and gas sector.

Variable-Level Discussion

1. Firm Size (FS)

Firm Size was found to have a positive and statistically significant relationship with ROA ($\beta = 0.647$, $p = 0.005$), suggesting that larger firms tend to achieve better financial outcomes. This supports existing literature which posits that firm size can enhance profitability through:

- Economies of scale and scope,
- Better access to financial and human capital,

- Stronger negotiation power in the market,
- More established corporate governance and operational structures.

The significance of FS in this context underscores the structural advantage that scale provides in the Nigerian oil and gas industry, which is capital-intensive and often subject to regulatory and infrastructural constraints.

2. Community Relation and Sustenance Disclosure (CRSD)

CRSD, which captures a firm's engagement in building long-term relationships with local communities, showed no significant effect on ROA ($p = 0.48$). Despite being a frequently disclosed aspect of CSR, the lack of measurable financial impact suggests that such initiatives may be implemented in a way that is either superficial or not fully integrated into the firm's strategic objectives. It may also reflect the long-term nature of community engagement, which may not yield immediate financial returns captured by short-term metrics like ROA.

3. Community Investment Disclosure (CID)

Similarly, CID was found to be statistically insignificant ($p = 0.79$). This implies that expenditures on community development projects—such as education, infrastructure, or social programs—do not have a direct correlation with profitability in the short term. This could be due to weak stakeholder perception, lack of alignment between community needs and corporate investments, or insufficient visibility and impact of such initiatives from a financial standpoint.

4. Health and Safety Program Disclosure (HSPD)

HSPD also did not show a significant relationship with ROA ($p = 0.92$). While health and safety practices are essential in the oil and gas sector for legal compliance and employee well-being, their

financial payoff may be indirect—such as in reduced accident costs, improved employee morale, or brand reputation—and thus not immediately evident in accounting returns. The insignificance could also point to standardization in disclosure across firms, limiting variability needed to detect financial effects.

Overall Implications

Although CSR disclosures are common among Nigerian oil and gas firms, their lack of significant influence on profitability suggests that these initiatives may currently serve symbolic or compliance functions rather than being strategically embedded in core business operations. The relatively low Adjusted R^2 (0.072) of the regression model further indicates that other factors—such as operational efficiency, market dynamics, governance structures, and macroeconomic conditions—are likely more influential in determining firm performance.

Moreover, the absence of multicollinearity, as confirmed by low correlation coefficients and VIF values, validates the model structure and confirms that the findings are not distorted by inter-variable dependencies.

Conclusion

The study's results highlight a critical gap between CSR disclosure practices and tangible financial outcomes in the Nigerian oil and gas industry. For CSR to become a true driver of performance, firms may need to transition from **declarative disclosures** to **strategically oriented CSR practices** that align with both stakeholder expectations and core business goals. Future research could expand the model to include moderating variables (e.g., corporate governance, environmental risk exposure) or investigate the **long-term effects** of CSR on performance using alternative metrics such as market-based indicators or stakeholder satisfaction.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study examined the impact of Corporate Social Responsibility Disclosure (CSRD) on the financial performance of selected Nigerian oil and gas companies from 2014 to 2023, using Return on Assets (ROA) as the key financial performance indicator. The CSRD components assessed included Community Relation and Sustenance Disclosure (CRSD), Community Investment Disclosure (CID), and Health and Safety Program Disclosure (HSPD), with Firm Size (FS) included as a control variable.

The findings revealed that among the independent variables, only Firm Size had a statistically significant positive effect on ROA, indicating that larger firms tend to report higher profitability. Conversely, the three CSRD variables (CRSD, CID, and HSPD) showed no significant impact on financial performance, suggesting that while these CSR disclosures are commonly reported, they do not directly enhance firm profitability in the short term. Descriptive statistics showed high mean values for the CSR variables, confirming widespread disclosure practices across firms. However, the regression analysis indicated that these disclosures may be more symbolic or reputational rather than financially driven.

Overall, the study provides evidence that firm size plays a more decisive role in determining financial performance than CSR disclosures, at least within the observed period and sector.

5.2 Conclusion

This study investigated the effect of Corporate Social Responsibility Disclosure (CSRD) on the financial performance of Nigerian oil and gas companies, using Return on Assets (ROA) as the

primary performance metric. The analysis focused on three key dimensions of CSRD—Community Relation and Sustenance Disclosure (CRSD), Community Investment Disclosure (CID), and Health and Safety Program Disclosure (HSPD)—while controlling for Firm Size (FS).

The findings revealed that CSRD, in its current form, does not have a statistically significant influence on firm profitability. Although CSR disclosures are generally prevalent among the sampled firms, their effect on financial performance was negligible. However, Firm Size was found to significantly and positively impact ROA, indicating that larger firms tend to achieve better financial outcomes. This implies that firm-specific characteristics such as size may play a more crucial role in financial performance than CSR-related initiatives alone.

In conclusion, while CSR disclosures are an essential part of corporate accountability and stakeholder engagement, their direct financial benefits may not be immediate or easily measurable through traditional accounting indicators like ROA. Therefore, companies may need to view CSR more as a long-term strategic investment rather than a short-term financial tool.

5.3 Recommendations

Based on the findings and conclusion of this study, the following recommendations are proposed:

1. **Strategic Integration of CSR into Core Business Operations:** Oil and gas companies should move beyond symbolic CSR reporting and integrate CSR activities into their long-term business strategies. This approach may enhance both stakeholder value and future financial performance.

2. **Improve the Quality and Relevance of CSR Disclosures:** Firms should focus on providing more detailed, transparent, and impactful CSR disclosures, particularly in areas

where disclosures currently show no measurable impact. Enhanced reporting practices may improve stakeholder trust and long-term sustainability.

3. **Scale CSR to Reflect Company Capacity:** Since firm size significantly influences financial performance, smaller firms should adopt CSR strategies that are proportionate to their resources, while larger firms should leverage their scale to implement more extensive and measurable CSR programs.

4. **Conduct Regular Impact Assessments:** Companies should periodically assess the actual outcomes of their CSR initiatives on financial and social performance to identify what works and ensure that CSR activities contribute meaningfully to business goals.

5. **Policy and Regulatory Support:** Regulatory bodies and industry stakeholders should establish clearer guidelines and frameworks for CSR reporting to ensure consistency, comparability, and alignment with sustainable development goals.

5.4 Suggestions for Further Studies

This study focused on the relationship between Corporate Social Responsibility Disclosure (CSRD) and financial performance, specifically Return on Assets (ROA), among selected Nigerian oil and gas firms over a ten-year period. While it has provided valuable insights, several opportunities exist for future research to build upon and extend the findings:

1. **Expand the Scope to Other Sectors**

Future studies could examine firms in other industries such as manufacturing, banking, or telecommunications to determine whether the observed relationships between CSRD and

financial performance are consistent across sectors with different regulatory, operational, and stakeholder dynamics.

2. **Incorporate Market-Based Performance Metrics**

This study employed an accounting-based performance measure (ROA). Future research could incorporate market-based indicators such as Tobin's Q, stock returns, or earnings per share (EPS) to capture broader and investor-driven dimensions of firm performance.

3. **Explore Longitudinal and Lag Effects**

The financial benefits of CSR initiatives may not be immediate. Future studies could use lagged variables or conduct longitudinal analysis to capture the delayed impact of CSR practices on financial performance over a longer time horizon.

4. **Include Moderating and Mediating Variables**

Future research could introduce potential mediators or moderators such as corporate governance mechanisms, board characteristics, environmental risks, or stakeholder engagement intensity to better understand the pathways through which CSR influences financial performance.

5. **Examine the Quality of CSR Disclosures**

Rather than focusing only on the quantity or presence of CSR disclosures, future studies could evaluate the **quality**, **credibility**, and **depth** of the disclosures using content analysis or third-party ratings to gain a more nuanced understanding of their financial relevance.

6. **Adopt Mixed-Methods Approaches**

Future researchers may consider combining quantitative analysis with qualitative techniques such as interviews, case studies, or focus groups to uncover managerial intentions, stakeholder perceptions, and contextual factors influencing CSR effectiveness.

7. **Investigate Regulatory and Institutional Influences**

Exploring how national policies, regulatory frameworks, or institutional pressures shape CSR practices and their performance outcomes can offer deeper insights into the institutional context of CSR in Nigeria and similar emerging economies.